

Practical multistate models

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Introduction

Multistate time to event models based on hazards, i.e., multistate hazard models (MSH) have been an extremely useful tool for the medical research teams that I have been involved in at Mayo Clinic. See for instance Jack et al, PMID:35310829 and PMID: 41240917, Allen et al PMID: 41240917 and PMID: 28941364 and Le-Rademacher PMID: 36003089 as examples. They are able to capture multiple aspects of a disease process, and thus can be an important tool for deeper understanding.

This document is a “nuts and bolts” description that reprises some of those analyses, with a focus on the data preparation, data checking, and analysis code needed. Some mathematics will appear, but only to help clarify the models.

For all the models, key first steps are

- What data do we have, and what questions *could* be answered with that data?
- The overall target: what aspects of the process(es) do we wish to describe, and in relation to what predictors? No paper can do everything.
- Lay out the multi-state model as a diagram. Refer to it often.
- Create the analysis data set, with particular attention to questions of immortal time bias.
- Look at the data (look, not just glance) and decide what to do with special cases. There are invariably one or two.

Mathematically we describe our multistate model in terms of three processes.

- $N_{ijk}(t)$ = the number of observed transitions from state j to state k , up to time t for subject i .
- $Y_{ij}(t)$ equals 1 if subject i is currently in state j , at time t , and at risk for transition to another state at that time. Subjects who have entered an absorbing state prior to t , e.g., death, are for instance not at risk for other transitions.
- $X_i(t)$ = the set of covariates for subject i at time t . Most often, most of these covariates will remain static, e.g., treatment arm or sex, while others are time-dependent, e.g. cholesterol might be remeasured at each visit.

Any careful investigation of the theory behind these models will need to distinguish between “at risk for an event” and “at risk for an observed event”. In this document I will largely ignore this, and use Y, N, X to refer to observed data. If censoring is not random, however, then the two notions are not the same, estimators can be biased, and any investigation into the hows and whys of that bias will need to carefully distinguish them.

Immortal Time Bias

In our opinion, one of the greatest threats to valid medical research is the presence of immortal time bias (ITB). In short, for a valid time-to-event model none of $N(t)$, $Y(t)$ nor $X(t)$ can contain values that depend on the future. MSH models, unfortunately, provide even more opportunities for this error to creep in, ones

which are sometimes difficult to spot. The number of ways to create ITB appears to be near infinite. Here are a few examples.

fill in later

Timeline data

The survival package supports two forms of multistate data. The older and better known one will be referred to as counting process form. The first 3 lines below show the myeloid data in its raw form as found in the survival package, followed by the counting process form. The tmerge call to create mdata1 will be discussed later.

```
library(survival)
library(mshmm)
table2 <- function(...) table(..., useNA="ifany")

# we will show this later
mdata1 <- tmerge(myeloid[,1:4], myeloid, id=id, death= event(futime, death),
                 CR = event(crttime), SCT= event(txtime), relapse=event(rltime),
                 priorcr = tdc(crttime), priorsct= tdc(txtime))

subset(myeloid, id<4)
```

```
##   id trt sex flt3 futime death txtime crttime rltime
## 1  1  B  f    C   235     1    NA     44    113
## 2  2  A  m    B   286     1   200     NA     NA
## 3  3  A  f    A  1983     0    NA     38     NA
```

```
subset(mdata1, id<4)
```

```
##   id trt sex flt3 tstart tstop death CR SCT relapse priorcr priorsct
## 1  1  B  f    C     0    44     0  1  0         0         0         0
## 2  1  B  f    C    44   113     0  0  0         1         1         0
## 3  1  B  f    C   113   235     1  0  0         0         1         0
## 4  2  A  m    B     0   200     0  0  1         0         0         0
## 5  2  A  m    B    200   286     1  0  0         0         0         1
## 6  3  A  f    A     0    38     0  1  0         0         0         0
## 7  3  A  f    A    38  1983     0  0  0         0         1         0
```

Each row of data in the timeline form consists of a subject id, a time interval (tstart, tstop], the event(s) that occur at the end of the interval (death, complete response, stem cell transplant, relapse), and covariates (trt, sex, flt3, priorcr, priorsct). Covariate apply over the entire interval (tstart, tstop]. Notice the difference between the event variable CR and the time-dependent covariate priorcr. Subject 1 experienced CR on day 44; the CR variable is 1 at the end of that first (0,44] interval and 0 thereafter. The time-dependent covariates priorcr becomes 1 at interval (44,113] and remains 1 thereafter. The fixed covariates of treatment, sex, and FLT3 repeat from row to row.

Now the timeline form of these 3 subjects.

```
subset(mdata2, id<4)
```

```
##   id time trt sex flt3 sct cr relapse death
## 1  1    0  B  f    C  NA  NA      NA    NA
## 2  1   44  B  f    C  NA  1      NA    NA
## 3  1  113  B  f    C  NA  NA      1    NA
## 4  1  235  B  f    C  NA  NA      NA     1
## 5  2    0  A  m    B  NA  NA      NA    NA
## 6  2   200 A  m    B   1  NA      NA    NA
```

```
## 7 2 286 A m B NA NA NA 1
## 8 3 0 A f A NA NA NA NA
## 9 3 38 A f A NA 1 NA NA
## 10 3 1983 A f A NA NA NA 0
```

In timeline data each row has a unique (subject identifier, time) pair, along with all the information that was observed at that time, with missing values for variables or outcomes that were not observed. Timeline data sets have many NA values. Static variables like treatment, sex and FLT3 state at randomization would normally be filled in for every row, as was done here. One could argue, however, that FLT3 was only observed at the start, since a pathology assessment was not repeated, so should be NA on row 2-4. Normally, most covariates will be non-missing at baseline, e.g., we know that CR, SCT, relapse and death have not yet occurred so would be 0. The NA in this case is a function of how we created the data, and will be handled properly by both Surv2 and lvcf since they are 0/1 (or FALSE/TRUE) variables.

For simple survival timeline style makes little sense: each subject will have to have two rows, one at 0 with the covariates and another at the death/censoring time with all covariates missing. For more multistate data there may be advantages.

- Counting process data sets often require the use of tmerge, which can be tricky. Timeline data can be built using more standard tools, which may be easier.
- There is only one CR and one SCT variable, not two of each. The choice between using a covariates as an endpoint or a time-dependent covariate is deferred until it actually appears in a model statement, on either the left hand side (as an event) or right hand side (as a covariate).
- If one is rigorous with respect to the “observed at this time” rule for each line of data, many immortal time bias problems will be avoided.

Example 1, NAFLD

The NAFLD data is found in 3 data sets. The original study contains all new NAFLD diagnosis in Olmsted County from 1997 to 2014, along with 4 potential controls for each case. Controls were selected matched on age, sex, and calendar year.

To protect patient confidentiality all time intervals are in days since the index date; none of the dates from the original data were retained. Subject age is their integer age at the index date, and the subject identifier is an arbitrary integer. As a final protection, the data includes only a 90% random sample of the original data.

- nafld1 has one row for each subject and contains the time fixed variables
- nafld2 contains time-dependent laboratory tests
- nafld3 contains the outcomes

```
nafld1[1:5,]
```

```
## id age male weight height bmi case.id futime status
## 1 1 57 0 60.0 163 22.69094 10630 6261 0
## 2 2 67 0 70.4 168 24.88403 14817 624 0
## 3 3 53 1 105.8 186 30.45354 3 1783 0
## 4 4 56 1 109.3 170 37.83010 6628 3143 0
## 5 5 68 1 NA NA NA 1871 1836 1
```

```
nafld2[1:5,]
```

```
## id days test value
## 1 1 -459 hdl 75
## 2 1 -459 chol 75
## 3 1 183 hdl 64
## 4 1 183 chol 64
## 5 1 2030 hdl 74
```

```
nafld3[1:5,]
```

```
##   id  days      event
## 1  3     0      nafld
## 2  4 -1287      htn
## 3  4 -1287 dyslipidemia
## 4  4 -1226    ang/isc
## 5  4   -90    ang/isc
```

```
table(nafld2$test)
```

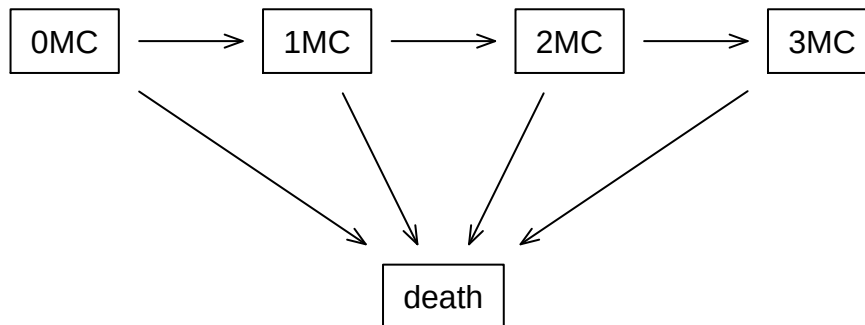
```
##
##   chol    dbp   fib4   hdl    sbp  smoke
## 161258 33430 3738 161259 33430 7008
```

```
table(nafld3$event)
```

```
##
##           afib           ang/isc cardiac arrest           diabetes heart failure
##           1935           2235           173           3452           1869
##           htn    dyslipidemia           MI           nafld           stroke
##           7097           10462           1199           3851           2054
```

The analysis in Allen 2018 looked at the accumulation of 3 metabolic comorbidities, leading to the 4 state model shown below. Age, male sex, NAFLD diagnosis, and BMI are the available baseline covariates. However, BMI is also a major risk factor for development of fatty liver. (We didn't include ongoing height/weight in the R data sets.)

```
nstate <- c("OMC", "1MC", "2MC", "3MC", "death")
nmat <- matrix(0, 5, 5, dimnames=list(nstate, nstate))
nmat[1,2] <- nmat[2,3] <- nmat[3,4] <- 1
nmat[1:4,5] <- 1
statefig(rbind(4,1), nmat)
```



In the `nafld1` data set, the `case.id` variable, for a control, gives the id of the case they were matched with; the incident cases have `case.id = id`. The `nafld3` data also has `nafld` diagnosis, so check for consistency.

```

ndata <- nafld1 # working data set
ndata$nafld <- 1*(ndata$id == ndata$case.id) # create a variable

table(ndata$nafld)

##
##      0      1
## 14004  3514

ndx <- subset(nafld3, event=="nafld")
table(ndx$days==0)

##
## FALSE  TRUE
##   318   3533

```

What is going on?

1. The 318 NAFLD diagnoses after day 0 are subjects who were chosen as controls, who were found to have NAFLD later in life. You cannot exclude someone as a control at age a based on something that will happen to them some years later at age $a + s$: this would be a form of immortal time bias. When that control later became a case, 4 new controls for them were selected.
2. At this time NAFLD had an interesting definition: if someone had fatty liver but also say a hepatitis C diagnosis, then they “didn’t have” NAFLD. The study question was thus more or less the consequences if NAFLD was the *only* liver disease.
3. What about hepatitis C 2 weeks later, 6 weeks later, ... The diagnostic process for someone with a GI

complaint can sometimes stretch out for months, so the decision was that “real” NAFLD starts once 12 months has passed without another smoking gun. This is day 0.

4. Controls need to be matched on this new date, otherwise the NAFLD subjects will benefit from immortal time bias.
5. The 19 who have NAFLD at day 0, but are not an incident case, are confusing. A of the 19 (not shown) reveals that all were originally chosen as a control. Perhaps I made a mistake when going from dates to days? (It should be “days from initial recruitment” for all).

“If you don’t know where you are going, you might end up somewhere else”. Yogi Berra

What should we do? Different choices lead to a different data set and analysis changes. * Simple approach: only follow the original enrollees and their controls. Ignore when someone changes the time-dependent covariate to NAFLD. * Treat NAFLD as a time-dependent covariate * Treat NAFLD as a time-dependent state * Restart: when a case becomes a control, split them into two copies: an original control and a new case.

Simple approach

For the simple analysis, we need a working data frame that has a row/days pair each time that the total number of MC increases. We can take advantage of the fact that each of the MC only is recorded at most once per subject: formally the variable is “history of diabetes”, etc. The lines below are likely an inefficient way to do this, but it works.

```
# first, subset to the relevant cases and controls
caseid <- subset(ndata, id==case.id)$id
tdata1 <- subset(ndata, case.id %in% caseid)
dim(tdata1) # this is the set subjects, one row each

## [1] 16253    10

# the endpoints of interest
enddata <- subset(nafl3, (id %in% tdata1$id) &
                  (event %in% c("diabetes", "dyslipidemia", "htn")))
enddata[1:10,]

##      id  days      event
## 2    4 -1287         htn
## 3    4 -1287 dyslipidemia
## 6    5   617         htn
## 7    7 -2685 dyslipidemia
## 8    9  -311      diabetes
## 10 10 -4498      diabetes
## 11 10 -4365 dyslipidemia
## 12 10 -3642         htn
## 13 11  -35         htn
## 14 12 -4723 dyslipidemia

table(enddata$days <=0)

##
## FALSE  TRUE
## 4644 14763

temp <- with(enddata, table(days <=0, event[,drop=TRUE]))
temp

##
##      diabetes  htn dyslipidemia
## FALSE      982 1964      1698
```

```
## TRUE 2219 4582 7962
```

```
round(temp/nrow(tdata1), 2)
```

```
##
## diabetes htn dyslipidemia
## FALSE 0.06 0.12 0.10
## TRUE 0.14 0.28 0.49
```

```
# counting process data
```

```
ndata2 <- tmerge(tdata1[c(1,2,3,6:7,10)], tdata1, id=id,
  death= event(futime, status),
  options=list(tstartname="day1", tstopname="day2"))
```

```
# Add the MC one by one
```

```
ndata2 <- tmerge(ndata2, subset(enddata, event=="diabetes"), id=id,
  diabetes= event(days), prior.diab= tdc(days))
ndata2 <- tmerge(ndata2, subset(enddata, event=="dyslipidemia"), id=id,
  dyslipid = event(days), prior.dysl= tdc(days))
ndata2 <- tmerge(ndata2, subset(enddata, event=="htn"), id=id,
  htn = event(days), prior.htn= tdc(days))
```

```
# alternate: add them all at once
```

```
ndata2b <- tmerge(ndata2, enddata, id=id, mcount=cumtdc(days),
  mevent= cumevent(days))
```

```
#
subset(ndata2, id <8)
```

```
## id age male bmi case.id nafld day1 day2 death diabetes prior.diab
## 1 1 57 0 22.69094 10630 0 0 6261 0 0 0
## 2 2 67 0 24.88403 14817 0 0 624 0 0 0
## 3 3 53 1 30.45354 3 1 0 1783 0 0 0
## 4 4 56 1 37.83010 6628 0 0 3143 0 0 0
## 5 5 68 1 NA 1871 0 0 617 0 0 0
## 6 5 68 1 NA 1871 0 617 1836 1 0 0
## 7 6 39 0 26.61559 7144 0 0 1581 0 0 0
## 8 7 49 0 25.51934 7507 0 0 3109 0 0 0
## dyslipid prior.dysl htn prior.htn
## 1 0 0 0 0
## 2 0 0 0 0
## 3 0 0 0 0
## 4 0 1 0 1
## 5 0 0 1 0
## 6 0 0 0 1
## 7 0 0 0 0
## 8 0 1 0 0
```

```
summary(ndata2) # a step by step summary of the additions
```

```
## Call:
```

```
## tmerge(data1 = ndata2, data2 = subset(enddata, event == "htn"),
## id = id, htn = event(days), prior.htn = tdc(days))
##
```

```
## early late gap within boundary leading trailing tied missid
## death 0 0 0 0 0 0 16253 0 0
## diabetes 2218 0 0 982 0 1 0 0 0
## prior.diab 2218 0 0 0 982 1 0 0 0
```

```
## dyslipid    7960    0    0   1659    37    2    2    0    0
## prior.dysl  7960    0    0    0   1696    2    2    0    0
## htn         4581    0    0   1893    66    1    5    0    0
## prior.htn   4581    0    0    0   1959    1    5    0    0
```

```
# Create the combined endpoint
nstate <- c("OMC", "1MC", "2MC", "3MC", "death")
temp1 <- with(ndata2, prior.diab + prior.dysl + prior.htn)
temp2 <- with(ndata2, diabetes + dyslipid + htn)
ndata2$istate <- factor(temp1, 0:3, nstate[1:4])
# temp3 = death(4), or on the day of a new MC, the new cumulative number
temp3 <- ifelse(ndata2$death==1, 4, ifelse(temp2==0, 0, temp2+temp1))
ndata2$event <- factor(temp3, 0:4, c("censor", nstate[-1]))
check1 <- survcheck(Surv(day1, day2, event) ~ 1, data=ndata2,
                     id=id, istate=istate)
check1
```

```
## Call:
## survcheck(formula = Surv(day1, day2, event) ~ 1, data = ndata2,
##           id = id, istate = istate)
##
```

```
## Unique identifiers      Observations      Transitions
##           16253           20787           5825
##
```

```
## Transitions table:
```

```
##           to
## from      1MC  2MC  3MC death (censored)
## OMC      1738   68    4   244         5304
## 1MC         0 1733   27   233         4231
## 2MC         0    0  970   397         3375
## 3MC         0    0    0   411         2052
## death      0    0    0    0            0
##
```

```
## Number of subjects with 0, 1, ... transitions to each state:
```

```
##           count
## state      0    1    2    3    4
## 1MC      14515 1738    0    0    0
## 2MC      14452 1801    0    0    0
## 3MC      15252 1001    0    0    0
## death    14968 1285    0    0    0
## (any)    11726 3446  884 177  20
```

```
# This won't be used further, but create the 9 state outcome of Diabetes,
# dyslipidemia=L and hypertension=H
```

```
temp1 <- with(ndata2, prior.diab + 2*prior.htn + 4*prior.dysl)
temp2 <- with(ndata2, diabetes + 2*htn + 4*dyslipid)
e1 <- with(ndata2, paste0(ifelse(prior.diab==1, "D", ""),
                           ifelse(prior.htn ==1, "H", ""),
                           ifelse(prior.dysl==1, "L", "")))
e1 <- ifelse(e1=="", "none", e1)
elab <- c("none", "D", "H", "L", "DH", "DL", "HL", "DHL") #preferred order
ndata2$e1 <- factor(e1, elab)

e2 <- ifelse(ndata2$death==1, 9, ifelse(temp2==0, 0, temp1+temp2))
ndata2$e2 <- factor(e2, c(0, 1,2,4,3,5,6,7,9), c("censor", elab[-1], "death"))
```

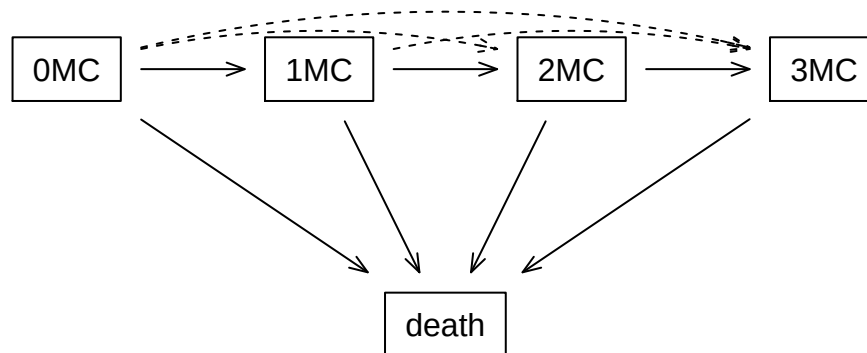
```
check2 <- survcheck(Surv(day1, day2, e2) ~ 1, ndata2, id=id, istate=e1)
```

For an MSH model fit with `survfit` or `coxph` the underlying likelihood is event based. Each observed event contributes a term to the likelihood, and presumes

- over the window of observation for a subject, all the transitions for that subject are recorded
- the window of observation is complete, i.e., it includes all times at which an event, if it occurred, would have been recorded
- non-informative censoring

One consequence of the first is that we do not need to inform the routine of which transitions can occur, and which are (effectively) absorbing: it tabulates them on its own. But, we do need to check that it's assessment agrees with our own. In this case we have some jumps that were not in our original concept.

```
nmat <- matrix(0, 5, 5, dimnames=list(nstate, nstate))
nmat[1,2] <- nmat[2,3] <- nmat[3,4] <- 1
nmat[1:4,5] <- 1
nmat[1, 3:4] <- nmat[2,4] <- .7
statefig(rbind(4,1), nmat, alty=c(1,2,1,2,2,1,1,1,1,1))
```



To properly account for age we will do the primary analysis on age scale.

```
quantile(tdata1$age)
```

```
##    0%   25%   50%   75%  100%
##    18    42    52    62    98
```

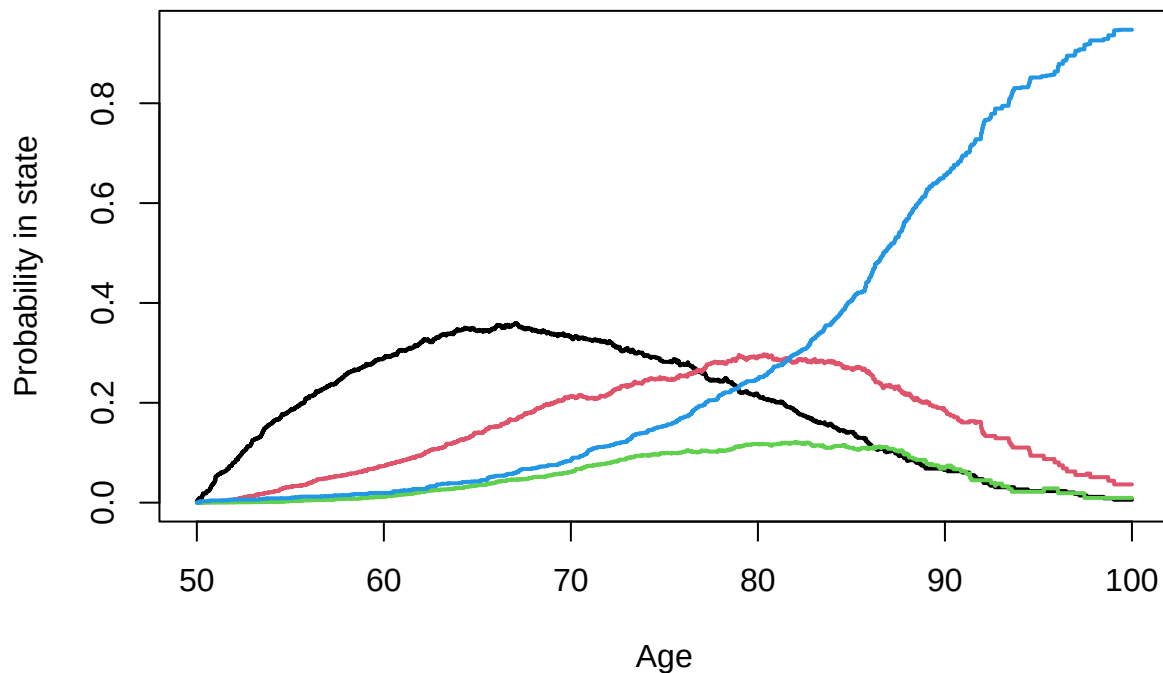
```
ndata2$age1 <- ndata2$age + ndata2$day1/365.25
ndata2$age2 <- ndata2$age + ndata2$day2/365.25
```

```
naj <- survfit(Surv(age1, age2, event) ~ male, ndata2, id=id, istate=istate,
               p0= c(1,0,0,0,0), start.time=50)

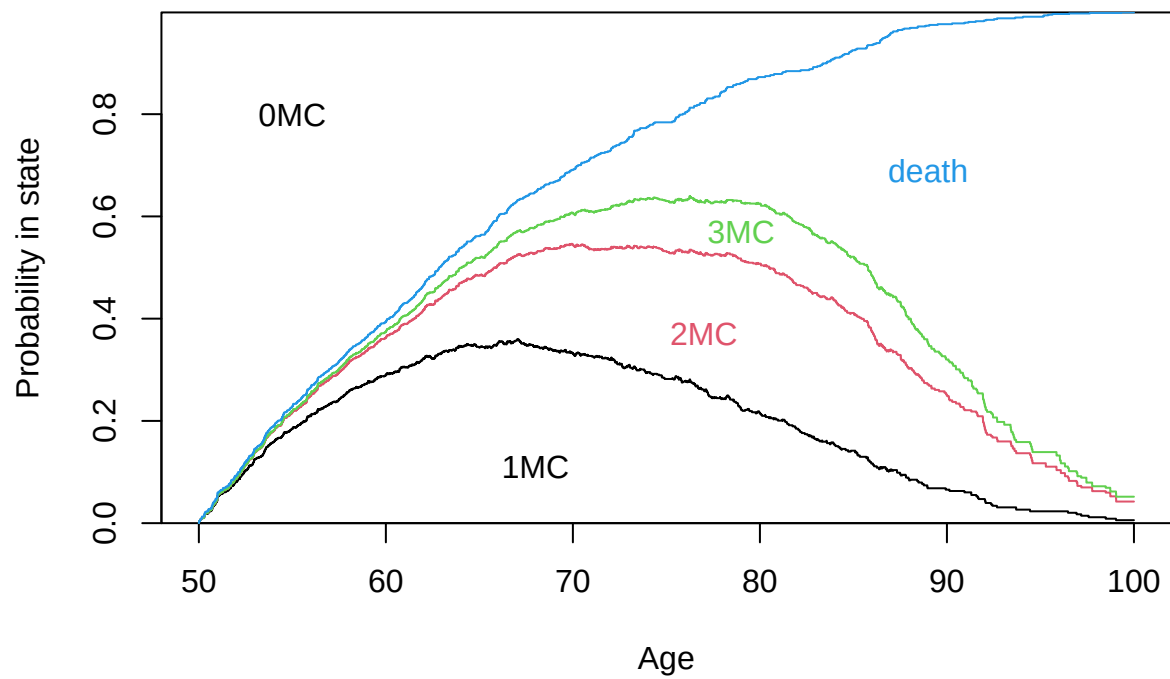
dim(naj)

## strata states
##      2      5

plot(naj[1,2:5], lwd=2, col=1:4, xmax=100,
     xlab="Age", ylab="Probability in state")
```



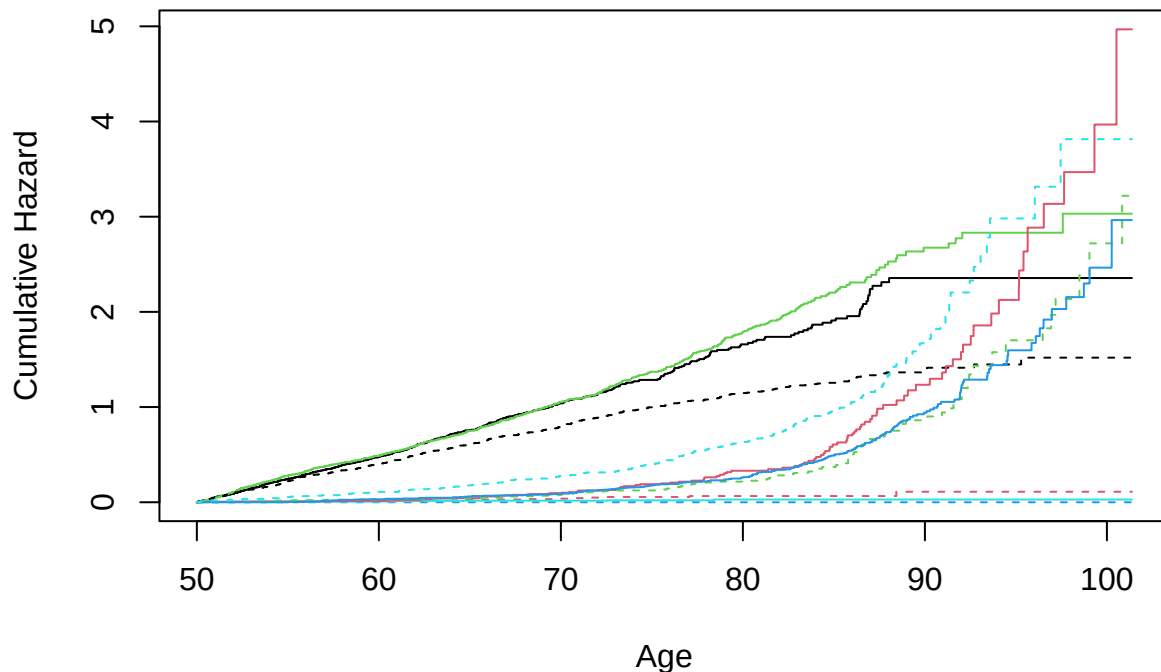
```
plot(naj[1,], cumprob=2:5, col=1:4, yaxs='i', xmax=100,
     xlab="Age", ylab="Probability in state")
#plot(naj, cumprob=2:5, lty=1:2, col=rep(1:4, each=2), yaxs='i', xmax=100,
#      xlab="Age", ylab="Probability in state")
text(c(55, 68, 77, 79, 89), c(.8, .11, .37, .57, .69), nstate,
     col=c(1,1,2,3,4))
```



```
dim(naj$cumhaz)
```

```
## [1] 13934 10
```

```
plot(naj[1,], cumhaz=TRUE, col=1:5, lty=1:2,
     xlab="Age", ylab="Cumulative Hazard")
```



The first two plots show probability in state, for females only, in two different forms. Some users prefer the first, some the second. The second figure is closely related to a stacked bar chart; the probability in state is the difference between two curves. The third plot shows the 10 separate estimated hazard curves. The probability in state estimates are mostly oblivious to the presence of 0mc:2MC, 0MC:3MC and 1MC:3MC transitions, but the hazard is estimated separately for each pair, some of which have small numbers. This affects the multi-state Cox model as well, since it is focused on hazards.

```
nacox1 <- coxph(Surv(age1, age2, event) ~ nafld + male,
                data=ndata2, id=id, istate=istate)
```

```
## Warning in agreg.fit(X, Y, istrat, offset, init, control, weights = weights, :
## Loglik converged before variable 8 ; beta may be infinite.
```

```
nacox1$cmap
```

```
##      1:2 1:3 2:3 1:4 2:4 3:4 1:5 2:5 3:5 4:5
## nafld   1   3   5   7   9  11  13  15  17  19
## male    2   4   6   8  10  12  14  16  18  20
```

The simple Cox model failed to converge. The coefficient map showed that coefficient 8 is associated with the 1:4 transition, which involves only a handful of subjects. To ameliorate this, bundle the coefficients for the small transitions.

```
nacox2 <- coxph(list(Surv(age1, age2, event) ~ nafld + male,
                    1:2 + 1:3 + 1:4 ~ nafld + male /common,
                    2:3 + 2:4 ~ nafld + male /common,
                    0:5 ~ male/common),
                data=ndata2, id=id, istate=istate)
```

```
nacox2
```

```
## Call:
## coxph(formula = list(Surv(age1, age2, event) ~ nafld + male,
##      1:2 + 1:3 + 1:4 ~ nafld + male/common, 2:3 + 2:4 ~ nafld +
##      male/common, 0:5 ~ male/common), data = ndata2, id = id,
##      istate = istate)
##
##
## 1:2, 1:3, 1:4      coef exp(coef) robust se      z      p
##      nafld 0.92706   2.52708   0.06932 13.374 < 2e-16
##      male  0.18656   1.20510   0.04940  3.776 0.000159
##
##
## 2:3, 2:4      coef exp(coef) robust se      z      p
##      nafld 0.52084   1.68344   0.05805  8.972 < 2e-16
##      male  0.21255   1.23683   0.05135  4.139 3.48e-05
##
##
## 3:4      coef exp(coef) robust se      z      p
##      nafld 0.48538   1.62479   0.07076  6.859 6.93e-12
##      male  0.11691   1.12401   0.06755  1.731 0.0835
##
##
## 1:5      coef exp(coef) robust se      z      p
##      nafld 0.68560   1.98497   0.23542  2.912 0.00359
##      male  0.33167   1.39329   0.05621  5.900 3.63e-09
##
##
## 2:5      coef exp(coef) robust se      z      p
##      nafld 0.53668   1.71033   0.15316  3.504 0.000458
##      male  0.33167   1.39329   0.05621  5.900 3.63e-09
##
##
## 3:5      coef exp(coef) robust se      z      p
##      nafld 0.51798   1.67863   0.10984  4.716 2.41e-06
##      male  0.33167   1.39329   0.05621  5.900 3.63e-09
##
##
## 4:5      coef exp(coef) robust se      z      p
##      nafld 0.03817   1.03891   0.10264  0.372 0.71
##      male  0.33167   1.39329   0.05621  5.900 3.63e-09
##
## States:  1= OMC, 2= 1MC, 3= 2MC, 4= 3MC, 5= death
##
## Likelihood ratio test=416.4  on 11 df, p=< 2.2e-16
## n= 20787, unique id= 16253, number of events= 5825
round(coef(nacox2, matrix=TRUE),3)
##      1:2  1:3  2:3  1:4  2:4  3:4  1:5  2:5  3:5  4:5
## nafld 0.927 0.927 0.521 0.927 0.521 0.485 0.686 0.537 0.518 0.038
## male  0.187 0.187 0.213 0.187 0.213 0.117 0.332 0.332 0.332 0.332
## attr("states")
## [1] "OMC" "1MC" "2MC" "3MC" "death"
```

Because all the covariates are time fixed, absolute risk estimates are fairly easy to get. For a Cox model,

predictions are available for any a priori combination of covariates, which we provide in a dummy data set. Since it is multistate, we need to give a distribution of starting states, and since we are on age scale a starting time. (If you don't give these it will guess. I don't think I have every found the result of that guess to be useful.)

```
dummy <- expand.grid(male=0:1, nafld=0:1)
coxsurv2 <- survfit(nacox2, newdata=dummy, p0=c(1,0,0,0,0), start.time=45)
dim(coxsurv2)

## data states
##      4      5

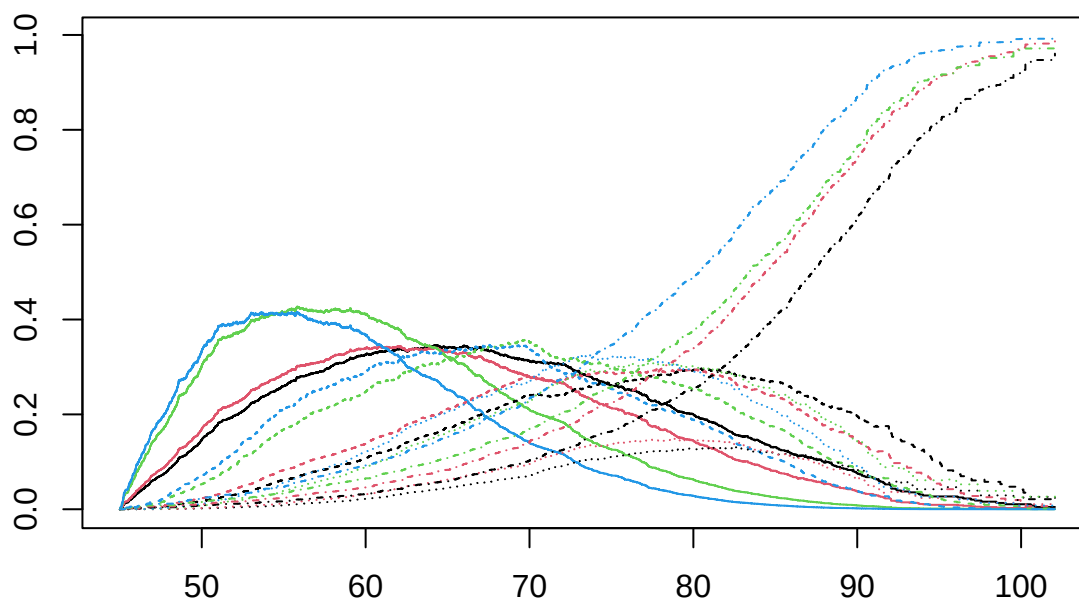
print(coxsurv2[1,], rmean=100)

## Call: survfit(formula = nacox2, newdata = dummy, start.time = 45, p0 = c(1,
##      0, 0, 0, 0))
##
##              n nevent      rmean*
## OMC      16919         0 17.881230
## 1MC      16919      1291 10.500605
## 2MC      16919      1541  8.656491
## 3MC      16919       899  3.238753
## death  16919      1254 14.722922
##      *restricted mean time in state (max time = 100 )

xx <- paste0("male=", dummy$male, ", Nafld=", dummy$nafld)
sojourn <- summary(coxsurv2, rmean=100)$table
sojourn <- matrix(sojourn[, "rmean"], ncol=4, byrow=TRUE,
                  dimnames=list(nstate, xx))
rbind(sojourn, Total= colSums(sojourn))

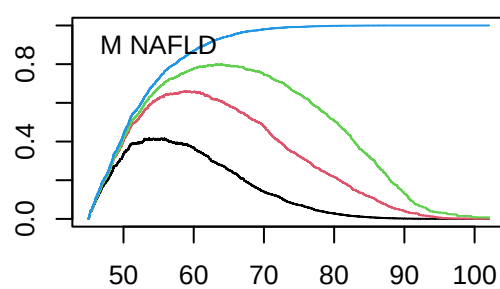
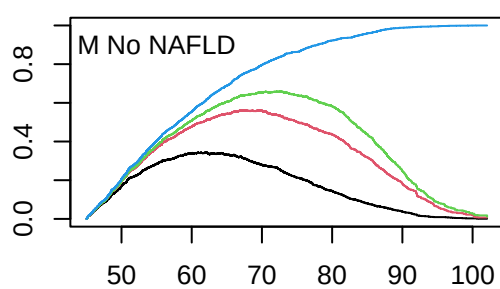
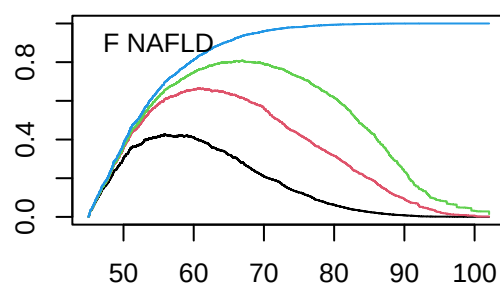
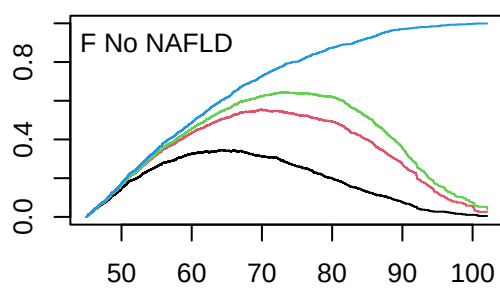
##      male=0, Nafld=0 male=1, Nafld=0 male=0, Nafld=1 male=1, Nafld=1
## OMC      17.881230      15.498161      8.885239      7.558078
## 1MC      10.500605      9.529256      9.558225      8.243057
## 2MC       8.656491      8.741058      9.801600      9.120698
## 3MC       3.238753      3.533408      7.997251      8.058475
## death    14.722922     17.698117     18.757685     22.019692
## Total    55.000000     55.000000     55.000000     55.000000

plot(coxsurv2[, -1], col=1:4, lty=rep(1:4, each=4))
```



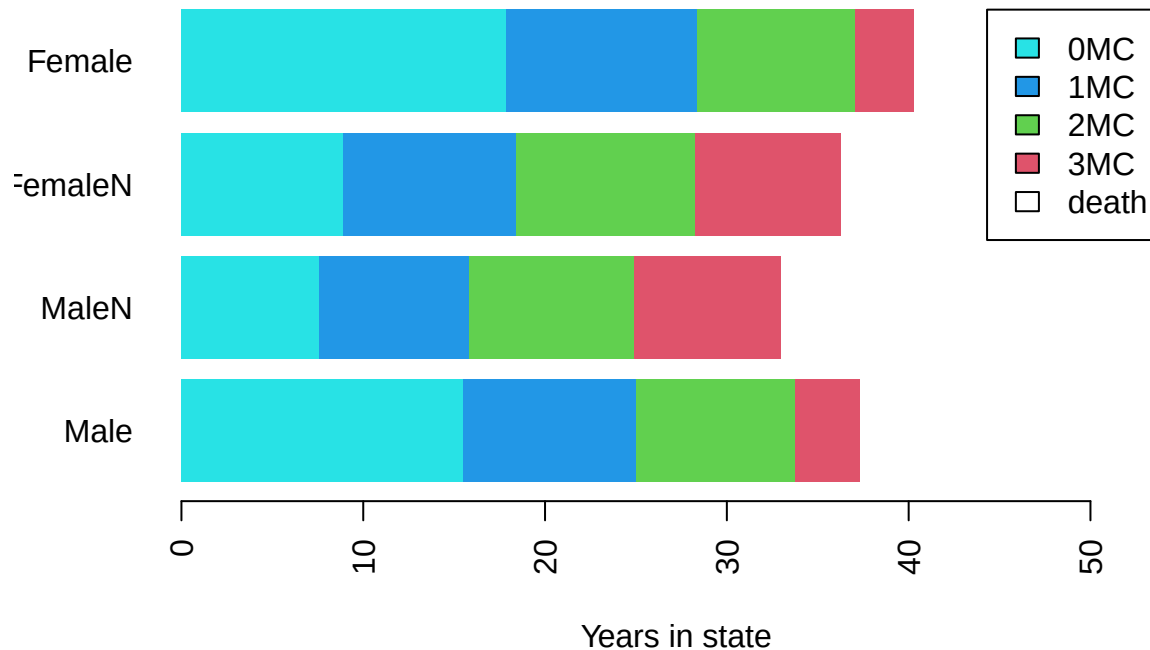
- The mean time in state has to add to $(100-45) = 55$, across the states.
- Subscripting the curves is a useful tool.
- The simple plot has too many curves, it looks like a bowl of spaghetti. Better plotting requires artistic choices.
- For the stratified model, the default plot will have a set of curves per stratum, 3854 times 5 states, which is obviously not useful.

```
opar <- par(mfcol=c(2,2), mar=c(5,5,2,1))
xx <- outer(c("F", "M"), c("No NAFLD", "NAFLD"), paste)
for (i in 1:4) {
  plot(coxsurv2[i,], cumprob=2:5, col=1:4, lty=rep(1:2, each=4))
  text(55, .9, xx[i])
}
```



```
par(opar)

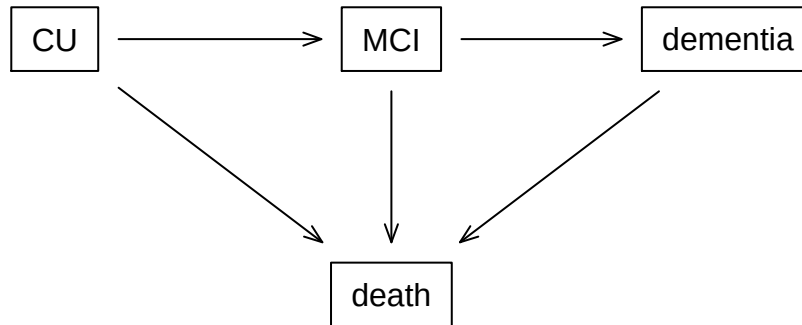
barplot(sojourn[,c(2,4,3,1)], horiz=TRUE, col=c(5:2, 0), las=2,
        names.arg=c("Male", "MaleN", "FemaleN", "Female"),
        xlab="Years in state", border=NA, legend=TRUE)
```



Mayo Clinic Study of Aging

The MCSA is based on a stratified random sample of subjects from Olmsted County, Minnesota. The dementia data set in the mshmm package is based on this data, and is able to reproduce Jack et al (2022) quite well, so we will use it for analysis. This data set is already in timeline form.

```
mstate <- c("CU", "MCI", "dementia", "death")
mmat <- matrix(0, 4, 4, dimnames=list(mstate, mstate))
mmat[1,2] <- mmat[2,3] <- 1
mmat[1:3,4] <- 1
statefig(rbind(3,1), mmat)
```



```
survcheck(Surv2(age, dx, repeated=TRUE) ~1, id=ptnum, dementia)$transitions
```

```
##           to
## from      CU  MCI dementia nondem off-dem death (censored)
## CU       13030 845      30    815    184    11      2234
## MCI       449 951     202    257    246    19      222
## dementia   0 29     163     0     0    172     30
## nondem     0 0      0      2     0    941     79
## off-dem    0 0      0      0     0    328    102
## death      0 0      0      0     0     0      0
```

```
survcheck(Surv2(age, dx) ~1, id=ptnum, data = dementia)
```

```
## Call:
## survcheck(formula = Surv2(age, dx) ~ 1, data = dementia, id = ptnum)
```

```
##
## Unique identifiers      Observations      Transitions
##           4317           21354           4528
##
```

```
## Transitions table:
```

```
##           to
## from      CU  MCI dementia nondem off-dem death (censored)
## CU         0 845      30    815    184    11      2347
## MCI       449 0     202    257    246    19      228
## dementia   0 29     0      0     0    172     30
## nondem     0 0      0      0     0    941     81
## off-dem    0 0      0      0     0    328    102
```

```
##   death      0    0      0      0      0    0      0
##
## Number of subjects with 0, 1, ... transitions to each state:
##           count
## state      0    1    2    3    4    5    6    7    9
##   CU      3912  363   40   2    0    0    0    0    0
##   MCI      3560  649  101   5    2    0    0    0    0
##   dementia 4094  214    9    0    0    0    0    0    0
##   nondem   3245 1072    0    0    0    0    0    0    0
##   off-dem   3887  430    0    0    0    0    0    0    0
##   death    2846 1471    0    0    0    0    0    0    0
##   (any)     2296  334 1125  380 123  47   9   2   1
```

We can see several things by looking at this table.

- The survcheck routine was written for counting process data. One aspect of Surv2, which translates from timeline form to that form, is the repeated argument. When a subject has two sequential visits in the same state, the survfit and coxph routines need to know whether to treat the second observation as a new event. Most of the time they should not: if “diabetes” is recorded in the medical history for 5 visits in a row this is not 5 new instances of the condition. Repeated infections in the CGD data set is a counterexample, however. In this data each row is a visit, the repeated form clearly shows that over half of all visit pairs are CU:CU; most patients stay stable for many years before progression. The (any) row of the total visits table shows that 53% (2296/4327) have no recorded transtions during follow-up.
- There are 30 instances of a CU:dementia pair. Per the model we assume that these subjects passed through MCI, we simply did assess them often enough to observe the transient MCI state.
- Dementia:MCI is much less common than MCI:CU. This reflects on one hand the clinical reality that CU and MCI are more difficult to distinguish than dementia, and also fewer overall dementia cases.
- There are no transitions from death to another state, or from death to censored (further follow-up after death). This is as it should be.
- Off study dementia only transitions to death. This is because further record reviews are not scheduled for demented subjects.
- There are no transitions from nondem to nondem or off-dem. This is, it turns out, an interesting artifact of how the online data collection system was designed. The nurse abstracters who review the history have a screen with fields for “last assessment date for dementia” and “dementia yes/no”. At a second, third, etc review for a patient, should one occur, the contents of these are overwritten; thus the master data set at any given time only has the most recent value.
- More dementias were found in record review than in study visits. We believe that this reflects informative censoring: subjects whose cognition is slipping selectively drop out, i.e., exactly those most likely to receive a dementia diagnosis at their next visit.

Jack et al (2022, PMID: 35310829) looked at a model for prediction of the absolute risk of dementia, a key question was the additional predictive power of amyloid PET after accounting for known risk factors of age, sex, the presence of one or more e4 alleles on the APOE gene, education, and CMC score (a count of 7 cardio-metabolic conditions). It used the modified endpoint of non-demented/dementia/death, collapsing the CU and MCI categories. First create the modified diagnosis variable:

```
dem <- dementia
table(dem$dx)

##
##   alive      CU      MCI dementia  nondem  off-dem  death
##   2680     17264     2357      395     1074     430    1471

temp <- c(1, 2,2,3,2,3,4)[as.numeric(dem$dx)]
dem$dx2 <- factor(temp, 1:4, c("censor", "ND", "dementia", "death"))
with(dem, table2(dx2, dx))
```

```
##          dx
## dx2      alive      CU      MCI dementia nondem off-dem death
##  censor    2680        0        0          0          0          0
##   ND        0 17264  2357          0    1074          0          0
##  dementia    0        0          0    395          0    430          0
##   death      0        0          0          0          0          0 1471
```

```
survcheck(Surv2(age, dx2) ~1, dem, id=ptnum)
```

```
## Call:
```

```
## survcheck(formula = Surv2(age, dx2) ~ 1, data = dem, id = ptnum)
```

```
##
```

```
## Unique identifiers      Observations      Transitions
```

```
##           4317           21354           2162
```

```
##
```

```
## Transitions table:
```

```
##          to
```

```
## from      ND dementia death (censored)
```

```
##   ND          0      662   971      2713
```

```
##  dementia   29          0   500      132
```

```
##   death      0          0    0          0
```

```
##
```

```
## Number of subjects with 0, 1, ... transitions to each state:
```

```
##          count
```

```
## state      0      1      2      3      4
```

```
##   ND      4288    29      0      0      0
```

```
##  dementia 3664   644      9      0      0
```

```
##   death   2846 1471      0      0      0
```

```
##   (any)   2707 1088 499 16      7
```

The above shows 29 transtions from dementia to ND. The 2022 analysis was actually for time to *first* dementia. This change to using only first event is common enough that we have added a routine to the survival package.

```
# the next line doesn't yet work though
# dx3 <- with(dem, nostutter(ptnum, dx, censor="alive", single=TRUE))
# do it with unlist(tapply(... which is very obscure
tfun <- function(x) { y <- as.numeric(x); ifelse(y==1, 1, cummax(y))}
temp <- with(dem, unlist(tapply(dx2, ptnum, tfun)))
```

```
dem$dx3 <- factor(temp, 1:4, c("censor", "ND", "dementia", "death"))
```

```
dem$edu4 <- pmax(8, dem$educ) /4
```

```
check2 <- survcheck(Surv2(age, dx3) ~1, dem, id=ptnum)
```

```
check2$transition
```

```
##
```

```
##          to
```

```
## from      dementia death (censored)
```

```
##   ND          653   957      2707
```

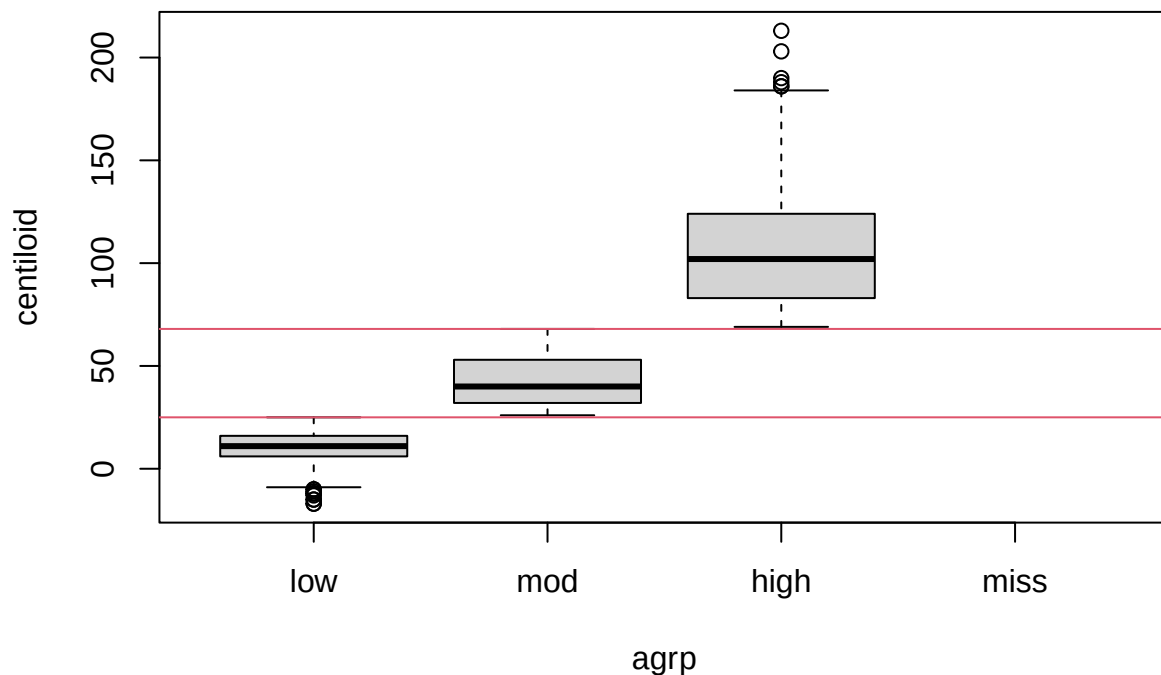
```
##  dementia      0   514      138
```

```
##   death      0      0          0
```

Amyloid PET raises some important statistical issues. First is that only about 40% of the subjects have amyloid imaging. This appears to be largely a time issue: imaging is a second appointment of 2 full days (MRI, amyloid PET, tau PET, optional FDG PET), and many who have previously given informed consent for imaging defer at this point. A second is that the ligand used for amyloid PET did not become available until several years after the start of the MCSA, so many of those who have imaging do not start at visit 1. Third is that the pattern of amyloid accumulation has an elbow, and each subject has a unique age for their bend. Some prior work had categorized amyloid as low/moderate/high; our choice in this study was to use a

time-dependent category of low/moderate/high/missing.

```
# add categorized amyloid, and initial CMC,
# but we want lvcf to work correctly wrt amyloid, apply it before making NA
# a category
atemp <- with(dem, lvcf(ptnum, centiloid))
temp <- ifelse(is.na(atemp), 4,
               as.numeric(cut(atemp, c(-50, 25, 68, 500))))
dem$agrp <- factor(temp, 1:4, c("low", "mod", "high", "miss"))
firstrow <- match(dem$ptnum, dem$ptnum)
dem$icmc <- dem$cmc[firstrow]
boxplot(centiloid ~ agrp, dem)
abline(h=c(25, 68), col=2)
```

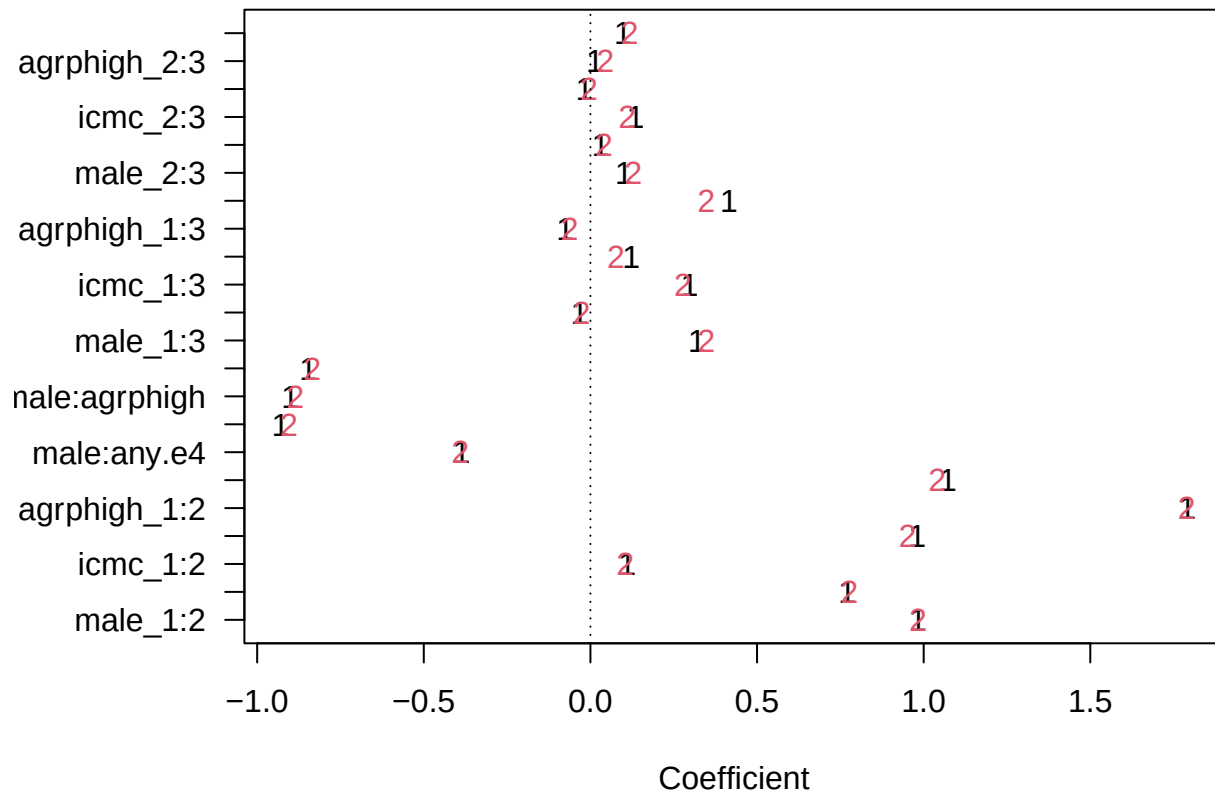


Now fit two models, using CMC and initial CMC.

```
mfit1 <- coxph(list(Surv2(age, dx3) ~ male + any.e4 + cmc + agrp,
                    1:2 ~ male*(any.e4 + agrp)),
               data= dem, id=ptnum)
mfit2 <- coxph(list(Surv2(age, dx3) ~ male + any.e4 + icmc + agrp,
                    1:2 ~ male*(any.e4 + agrp)),
               data= dem, id=ptnum)

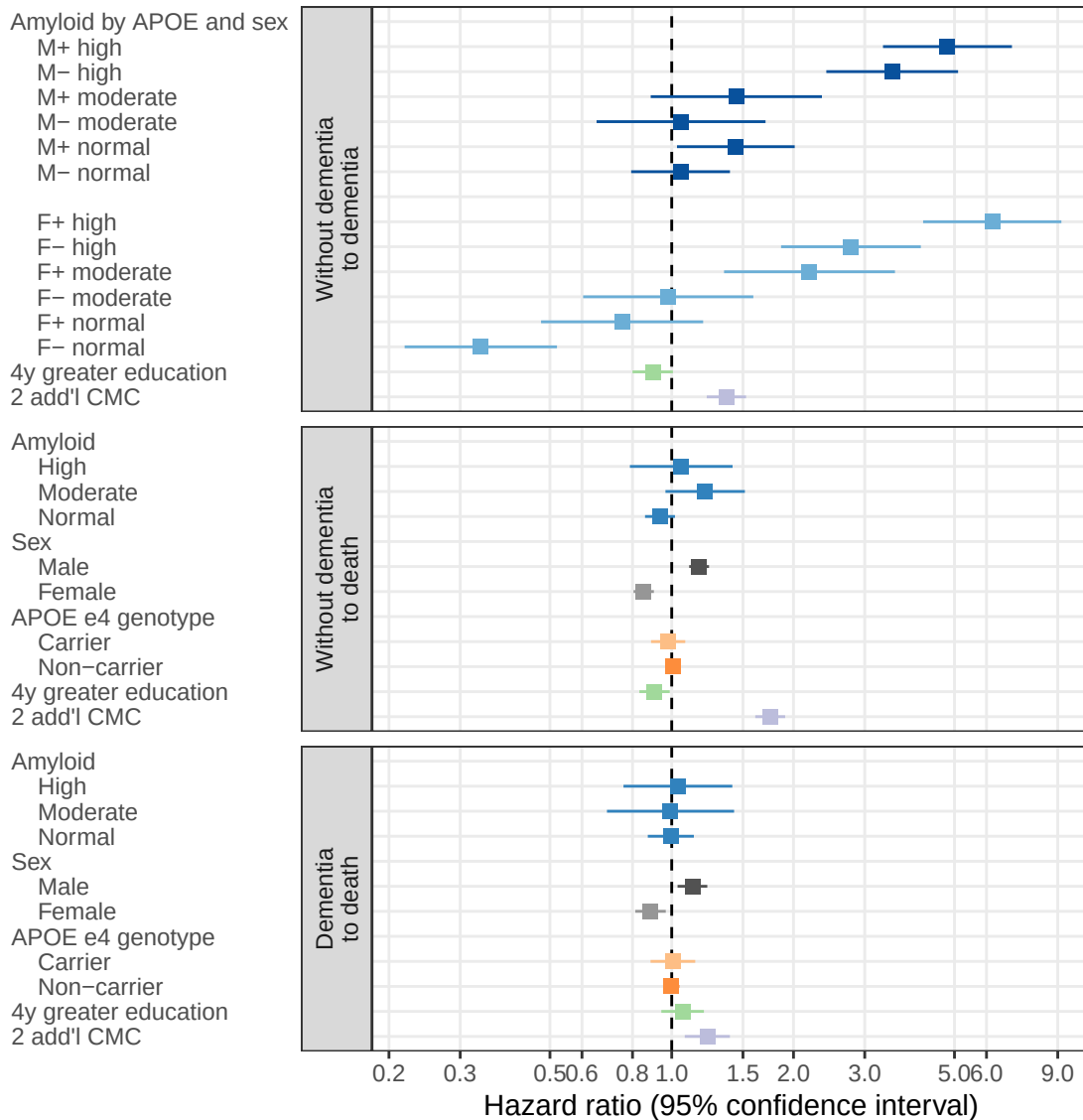
opar <- par(mar=c(5,6,1,1))
matplot(cbind(coef(mfit1), coef(mfit2)), 1:22, xlab="Coefficient",
         ylab="", yaxt='n')
abline(v=0, lty=3)
```

```
axis(2, 1:22, names(coef(mfit2)), las=2)
```



```
par(opar)
```

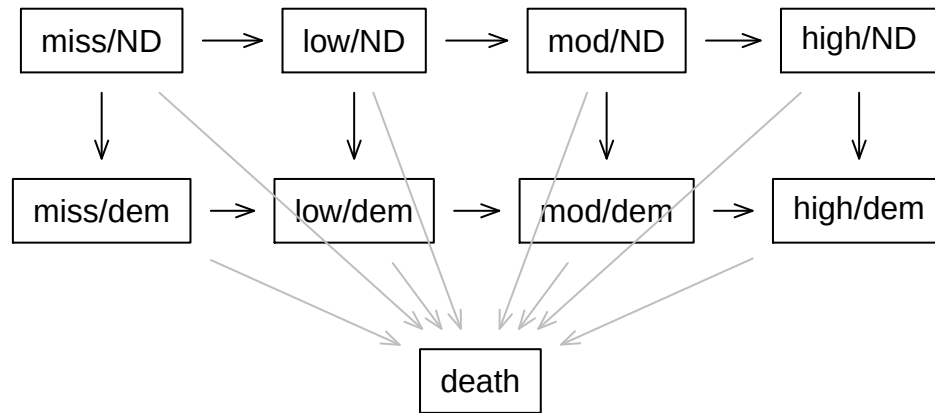
Not done: create my own version of Emily's nice plot (below) and insert it instead.



How do I get a probability in state curve for a 65 year old female, e4-, initial CMC of 2, current state of ND, and *current* centiloid of 30, when centiloid is a time dependent covariate? The standard survfit call (curves for a coxph model) doesn't work for TD covariates. For CMC, we are able to dodge the issue by instead using a time-fixed variable of initial CMC at recruitment. The plot above shows that hazard ratios don't change very much. But that is not an option for amyloid: any "initial amyloid" variable that we try to create leads to immortal time bias.

One approach is to make more states, creating 4 amyloid * (ND, dementia) + death = 9 states. One could also have 4*3 = 12, separating out death with low amyloid, death with moderate amyloid, etc. (The reason to use initial CMC instead of time-dependent CMC is so that this example has a more manageable number of states. The plot above shows that the models with initial and time dependent CMC are quite similar.)

Create this new set of states. Code wise, it was easiest to first fill out the censored values in the `dx3` variable to make it complete. We also need to create the first dementia status ourselves, as the automated methods may not recognize a mod/dementia to mod/ND transition as a should-not-be-allowed repeat of ND.



```
with(dem, table(agrp, dx3))
```

```
##          dx3
## agrp   censor   ND dementia death
##  low      871  5185      71   138
##  mod      182  1202      44    68
##  high     154   979     130    99
##  miss    1473 13290     619  1166
```

```
temp1 <- with(dem, ifelse(dx3=="censor", NA, as.numeric(dx3)))
temp2 <- with(dem, lvcf(ptnum, temp1))
temp2 <- factor(temp2, 2:4, c("ND", "dementia", "death"))
with(dem, table2(agrp, temp2))
```

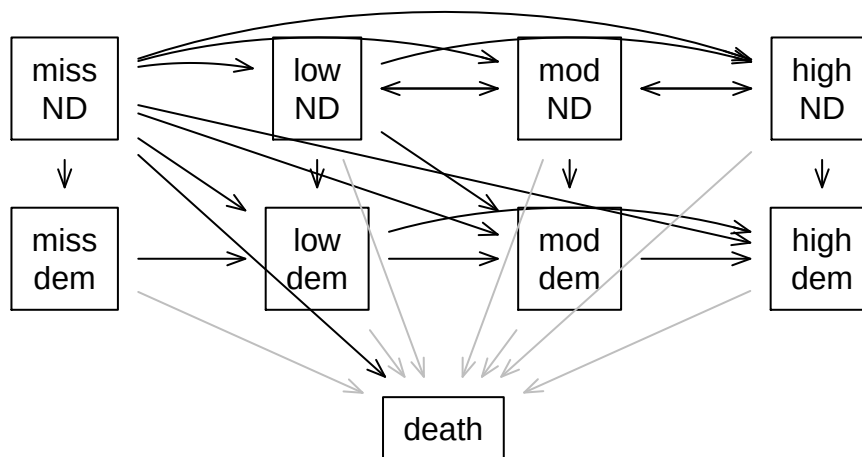
```
##          temp2
## agrp      ND dementia death
##  low    6039      88   138
##  mod    1373      55    68
##  high   1103     160    99
##  miss 14683     699  1166
```

```
temp3 <- with(dem, as.numeric(agrp) + 4* as.numeric(temp2) -4)
# the pmin collapses 9-12= low/death, ..., miss/death in to "death"
dem$dx4 <- factor(pmin(9, temp3), 0:9,
  c("censor",
    outer(c("low", "mod", "high", "miss"), c("ND", "dem"),
      paste, sep='/'), "death"))
```

```
check3 <- survcheck(Surv2(age, dx4) ~ 1, id=ptnum, dem)
check3$trans
```

```
##           to
## from      low/ND mod/ND high/ND low/dem mod/dem high/dem miss/dem death
## low/ND      0   107     2    49     1     0     0     0   104
## mod/ND      9     0    65     0    31     0     0     0    42
## high/ND     0     1     0     0     0    86     0     0    39
## miss/ND    570   130   144     1     3     5    477   772
## low/dem     0     0     0     0     1     0     0     0    34
## mod/dem     0     0     0     0     0     0     0     0    26
## high/dem    0     0     0     0     0     0     0     0    60
## miss/dem    0     0     0     2     1     0     0     0   394
## death      0     0     0     0     0     0     0     0     0

##           to
## from      (censored)
## low/ND      908
## mod/ND      178
## high/ND      137
## miss/ND     1476
## low/dem      17
## mod/dem       11
## high/dem     30
## miss/dem     80
## death        0
```



Writing a coxph call is going to be a bother: there are 9 transitions that have a count of 5 or less, all of which

would need to be bundled with some other one(s) to avoid undefined coefficients. Take another approach. Start with the simpler figure.

For the MSH model, I am interested in only the downward arrows. One could explore the other factors, if any, the affect amyloid progression, i.e., rightward arrows from low to moderate or moderate to high, but the focus of the 2022 paper was on the prediction of cognitive state. Transitions from missing amyloid to low/moderate/high amyloid, lines 4 and 8 of the transitions table, occur for subjects who entered the study before the PET ligand for amyloid (Pittsburg compound B) became available, those transitions are based on calendar date of the visit and have nothing to do with disease per se.

An alias data set is useful here; it allows shorthand formulas within the coxph and cmsh functions.

alias

```
##      state A C
## 1 miss/ND 1 1
## 2 low/ND 2 1
## 3 mod/ND 3 1
## 4 high/ND 4 1
## 5 miss/dem 1 2
## 6 low/dem 2 2
## 7 mod/dem 3 2
## 8 high/dem 4 2
## 9 death 0 3
```

```
mfit3 <- coxph(list(Surv2(age, dx4) ~ 1 ,
                    C(1):C(2) ~ male*(any.e4 + agrp) + icmc + 1/ common,
                    C(1):"death" ~ male + any.e4 + icmc + agrp + 1 /common,
                    C(2):"death" ~ male + any.e4 + icmc + agrp + 1 /common),
                data=dem, id=ptnum, statedata= alias)
```

mfit3

Call:

```
## coxph(formula = list(Surv2(age, dx4) ~ 1, C(1):C(2) ~ male *
## (any.e4 + agrp) + icmc + 1/common, C(1):"death" ~ male +
## any.e4 + icmc + agrp + 1/common, C(2):"death" ~ male + any.e4 +
## icmc + agrp + 1/common), data = dem, id = ptnum, statedata = alias)
```

##

##

## 1:5, 4:5, 1:6, 2:6, 4:6, 3:7, 4:7, 4:8	coef	exp(coef)	robust se	z
## male	0.98314	2.67285	0.33173	2.964
## any.e4	0.77700	2.17494	0.11467	6.776
## agrpmod	0.95262	2.59250	0.42070	2.264
## agrphigh	1.78890	5.98286	0.32911	5.436
## agrpmiss	1.04152	2.83353	0.29662	3.511
## icmc	0.10507	1.11079	0.03072	3.420
## male:any.e4	-0.38998	0.67707	0.16410	-2.376
## male:agrpmod	-0.90396	0.40496	0.51133	-1.768
## male:agrphigh	-0.88564	0.41245	0.40160	-2.205
## male:agrpmiss	-0.83388	0.43436	0.34627	-2.408

##

## 1:5, 4:5, 1:6, 2:6, 4:6, 3:7, 4:7, 4:8	p
## male	0.003040
## any.e4	1.24e-11
## agrpmod	0.023551
## agrphigh	5.46e-08
## agrpmiss	0.000446

```
##                                icmc            0.000626
##                                male:any.e4      0.017481
##                                male:agrpmod      0.077085
##                                male:agrphigh     0.027434
##                                male:agrpmiss     0.016033
##
##
## 1:9, 2:9, 3:9, 4:9      coef exp(coef) robust se      z      p
##      male      0.34852   1.41696   0.06588   5.290 1.22e-07
##      any.e4    -0.02614   0.97420   0.07477  -0.350 0.72665
##      agrpmod    0.07740   1.08047   0.18317   0.423 0.67264
##      agrphigh  -0.06180   0.94008   0.18761  -0.329 0.74187
##      agrpmiss   0.34757   1.41562   0.10659   3.261 0.00111
##      icmc       0.27781   1.32024   0.02516  11.041 < 2e-16
##
##
## 5:9, 6:9, 7:9, 8:9      coef exp(coef) robust se      z      p
##      male      0.129055   1.137753   0.086707   1.488 0.13665
##      any.e4     0.040780   1.041623   0.086602   0.471 0.63772
##      agrpmod   -0.003915   0.996093   0.245794  -0.016 0.98729
##      agrphigh   0.044825   1.045845   0.200387   0.224 0.82300
##      agrpmiss   0.118365   1.125655   0.173960   0.680 0.49624
##      icmc       0.110765   1.117132   0.034346   3.225 0.00126
##
## States: 1= low/ND, 2= mod/ND, 3= high/ND, 4= miss/ND, 5= low/dem, 6= mod/dem, 7= high/dem, 8= miss/ND
##
## Likelihood ratio test=351.9 on 22 df, p=< 2.2e-16
## n= 21354, unique id= 4317, number of events= 2123
```

In a Cox model, either single endpoint or a MSH, the baseline hazard plays a similar role to the intercept in a GLM, but its estimation can be deferred until after solving for the coefficients β ; it is similar in that way to the estimation of $\hat{\sigma}$ in a least-squared model. In the code above the combination of `agrp + 1` and `\common` on line 1 of the formula signals that proportional baseline hazards will be used for the set of ND:dementia transitions, i.e. assume hazards of

$$\begin{aligned} \lambda_{ND,dem}(t) \\ \lambda_{ND,dem}(t)e^{\gamma_1} \\ \lambda_{ND,dem}(t)e^{\gamma_2} \\ \lambda_{ND,dem}(t)e^{\gamma_3} \end{aligned}$$

for transitions from the low/ND to any of the 4 dementia states, for transitions from mod/ND (γ_1) to any of the 4 dementia states, for transitions from high/ND (γ_2) and for transitions from miss/ND (γ_3), respectively. (The γ s are in the order of the factor levels for `agrp`, not the order of my picture). This includes for instance the one diagonal transition from low/ND to mod/dem found in the transitions table. These three γ coefficients appear as `agrp` in the first block of printed coefficients for the model.

In `mfit3`, amyloid is no longer a time dependent variable but rather a state. There in fact no time-dependent coefficients, and so we are able to create honest predictions based on baseline covariates, starting age, and starting state.

```
dummy1 <- expand.grid(male=0:1, any.e4 = 0:1, icmc=1)

#msurv3.1 <- survfit(mfit3, newdata = dummy1, start.time=65,
#                    p0=c(1,0,0,0,0,0,0,0))
# the above call gives an error
# this is a flaw in the survival package, which is being addressed
```

What is most interesting is that the log-likelihood and coefficients from `mfit2` and `mfit3` are identical. This is because the underlying partial likelihood terms, one for each transition with covariates, have exactly the same set of subjects at risk, with the same covariate pattern. The coefficient's names and order differ, but values agree.

```
rbind(mfit2= mfit2$loglik, mfit3= mfit3$loglik)
```

```
##           [,1]      [,2]
## mfit2 -12811.63 -12635.7
## mfit3 -12811.63 -12635.7
```

One of the challenging aspects of multistate models is managing the total number of parameters. There are 35 separate transitions in this example, the number of non-zero elements in the transitions matrix. Without constraints, each transition will be assigned its own coefficient values for male, any.e4, cmc and amyloid, $6 * 35 = 210$ coefficients, along with 35 separate baseline hazard functions. One way to think of this is that the `coxph` function creates transition * covariate interactions by default; the same combinatorial explosion would occur in linear models if we included all possible interactions.